SMART DISASTER RELIEF INVENTORY AND DISTRIBUTION MANAGEMENT SYSTEM USING C

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ABSTRACT

Natural disasters such as floods, cyclones, earthquakes, landslides, and tsunamis can cause significant damage to human life, property, and infrastructure. During such emergency situations, the proper management and timely distribution of essential resources such as food packets, drinking water, medicines, blankets, rescue equipment, and temporary shelters are extremely important. Manual management of disaster relief resources may lead to inaccurate records, duplicate entries, resource shortages, delayed distribution, and loss of important information.

The Smart Disaster Relief Inventory and Distribution Management System using C is a menu-driven application developed to manage and monitor disaster relief resources efficiently. The system allows administrators to add, update, delete, search, sort, merge, analyse, distribute, and permanently store resource information. Each resource record contains details such as Resource ID, Resource Name, Category, Quantity Available, Minimum Required Quantity, Distribution Centre, Priority Level, and Distribution Demand.

The application uses arrays and strings to maintain resource records and supports searching and sorting based on different criteria such as quantity, priority, category, and distribution centre. The system also merges resource records received from multiple relief centres while avoiding duplicate Resource IDs. User-defined functions are used to make the program modular and scalable. Pointers and recursion are implemented for resource searching and quantity analysis.

File handling is used to permanently store resource information and generate disaster-resource reports. The system analyses the availability of resources and classifies them as Adequate, Low, or Critical based on predefined thresholds. The project demonstrates practical application of operators, decision-making statements, loops, arrays, strings, structures, searching, sorting, merging, functions, pointers, recursion, and file handling.

Keywords: Disaster Management, Relief Resources, Inventory Management, Resource Distribution, C Programming, File Handling, Recursion, Pointers, Searching and Sorting.

INTRODUCTION

A disaster is a sudden event that causes serious disruption to normal life and results in significant damage to people, property, infrastructure, and the environment. Disasters may occur naturally, such as floods, cyclones, earthquakes, tsunamis, landslides, and droughts, or they may result from human activities such as industrial accidents, fires, and chemical leaks.

During disaster situations, affected people often lose access to essential requirements such as food, drinking water, medicines, shelter, clothing, and emergency medical support. Disaster-management authorities must therefore ensure that relief resources are available in sufficient quantities and distributed efficiently to affected areas.

Managing disaster relief resources is a challenging task because multiple distribution centres may be involved, resource quantities may change continuously, and shortages must be identified quickly. Traditional manual methods can result in errors, duplicate entries, delays, and difficulty in tracking available stock.

The Smart Disaster Relief Inventory and Distribution Management System using C is designed to provide a simple and efficient solution for managing disaster relief resources. The system maintains information about different resources, including their unique identification number, name, category, available quantity, minimum required quantity, distribution centre, priority level, and demand.

The project uses C programming concepts such as arrays, structures, strings, functions, pointers, recursion, searching, sorting, and file handling to develop a modular and scalable application.

1. PROBLEM STATEMENT

Natural disasters such as floods, cyclones, earthquakes, tsunamis, and landslides create an immediate requirement for essential resources including food packets, drinking water, medicines, blankets, rescue equipment, and temporary shelters. During such situations, disaster-management authorities must efficiently monitor available resources and distribute them to affected people.

Manual management of relief resources can lead to duplicate records, inaccurate stock information, delayed identification of shortages, inefficient distribution, and loss of important data.

The proposed system aims to design and implement a menu-driven C application for managing disaster relief resources. The application allows administrators to add, update, delete, search, sort, merge, analyse, distribute, and store relief-resource information.

The system maintains the following details:

- Resource ID
- Resource Name
- Category
- Quantity Available
- Minimum Required Quantity
- Distribution Centre
- Priority Level
- Distribution Demand

The application identifies duplicate Resource IDs while merging records from multiple relief centres. It analyses resource availability and classifies resources as Adequate, Low, or Critical. The program also generates consolidated reports showing total available resources, critical resources, immediate replenishment requirements, and resources with high distribution demand.

2. OBJECTIVE

2.1 Primary Objective

To develop a Smart Disaster Relief Inventory and Distribution Management System using C programming for efficient monitoring and distribution of essential resources during disaster situations.

2.2 Specific Objectives

1. To maintain records of disaster relief resources.
2. To add new resource information.
3. To update existing resource details.
4. To delete unnecessary records.
5. To search resources using Resource ID.
6. To search resources based on category.
7. To sort resources based on quantity.
8. To sort resources based on priority.
9. To sort resources based on distribution centre.
10. To merge resource records from multiple centres.
11. To prevent duplicate Resource IDs.
12. To analyse resource availability.
13. To classify resources as Adequate, Low, or Critical.
14. To identify resources requiring immediate replenishment.
15. To distribute resources based on availability.
16. To calculate total available resources.
17. To demonstrate pointers and recursion.
18. To store data permanently using file handling.
19. To generate disaster resource reports.

3. REQUIREMENTS AND ENVIRONMENT USED

3.1 Hardware Requirements

Component	Requirement
Processor	Intel Core i3 or Above
RAM	Minimum 4 GB
Storage	500 MB Free Space
Monitor	Standard Display
Input Device	Keyboard

3.2 Software Requirements

Software	Requirement
Operating System	Windows / Linux
Programming Language	C
Compiler	GCC
IDE	CodeBlocks / Dev C++ / VS Code
File Storage	Local File System

3.3 Programming Concepts Used

- Operators
- Decision-making statements
- Switch-case
- Loops
- Arrays
- Strings
- Structures
- Functions
- Pointers
- Recursion
- Searching
- Sorting
- Merging
- File Handling

4. DESIGN / PROPOSED SOLUTION

The proposed system is a menu-driven C application designed to manage disaster relief resources efficiently. The administrator interacts with the system through a main menu and can perform different operations.

4.1 System Architecture

ADMIN → MAIN MENU SYSTEM → RESOURCE MANAGEMENT / SEARCHING & SORTING / RESOURCE ANALYSIS / FILE HANDLING → DISASTER RESOURCE REPORT GENERATION

4.2 Modules of the System

Resource Management Module

- Add Resource
- Display Resource
- Update Resource
- Delete Resource

Searching Module

- Search by Resource ID
- Search by Category

Sorting Module

- Sort by Quantity
- Sort by Priority
- Sort by Distribution Centre

Analysis Module

Condition	Status
Quantity ≥ 2 × Minimum Required	Adequate
Quantity ≥ Minimum Required and < 2 × Minimum Required	Low
Quantity < Minimum Required	Critical

Distribution Module

Checks available stock before distributing resources and prevents allocation beyond available quantity.

Merge Module

Merges records from multiple relief centres while preventing duplicate Resource IDs.

File Handling Module

- Save Data
- Load Data
- Generate Report

5. ALGORITHM / PSEUDOCODE / FLOWCHART

5.1 Main Algorithm

20. Start the program.
21. Load previously saved data.
22. Display the main menu.
23. Read user choice.
24. Execute the selected operation.
25. Repeat until the Exit option is selected.
26. Save data.
27. Stop.

5.2 Pseudocode

```
START

Load Resource Data

DO
    Display Menu
    Read Choice

    SWITCH Choice
        CASE 1: Add Resource
        CASE 2: Display Resources
        CASE 3: Update Resource
        CASE 4: Search Resource
        CASE 5: Search by Category
        CASE 6: Sort by Quantity
        CASE 7: Sort by Priority
        CASE 8: Sort by Centre
        CASE 9: Analyse Resources
        CASE 10: Distribute Resource
        CASE 11: Merge Resources
        CASE 12: Delete Resource
        CASE 13: Save Data
```

```
        CASE 14: Load Data
        CASE 15: Generate Report
        CASE 0: Exit
        DEFAULT: Display Invalid Choice
    END SWITCH

WHILE Choice != 0

Save Data
STOP
```

5.3 Flowchart

```
START
  ↓
Load Saved Data
  ↓
Display Main Menu
  ↓
Read Choice
  ↓
Perform Selected Operation
  ↓
Exit?
  ├── No → Return to Main Menu
  └── Yes → Save Data → STOP
```

6. IMPLEMENTATION / SOURCE CODE

The system is implemented using an array of structures, functions, pointers, recursion, searching, sorting, merging, and file handling.

6.1 Structure Used

```
typedef struct {
    int resourceID;
    char resourceName[50];
    char category[30];
    int quantityAvailable;
    int minimumRequired;
    char distributionCentre[50];
    int priority;
    int distributionDemand;
} Resource;typedef struct {
    int resourceID;
    char resourceName[50];
    char category[30];
    int quantityAvailable;
    int minimumRequired;
    char distributionCentre[50];
    int priority;
    int distributionDemand;
} Resource;
```

6.2 Major Functionalities

- addResource()
- displayResources()
- updateResource()
- searchResource()
- searchByCategory()
- sortByQuantity()
- sortByPriority()
- sortByCentre()
- analyseResources()
- distributeResource()

- mergeResources()
- deleteResource()
- saveToFile()
- loadFromFile()
- generateReport()
- recursiveTotalQuantity()
- recursiveCriticalResources()

7. TEST CASES AND EXPECTED / ACTUAL RESULTS

Test Case	Input	Expected Result	Actual Result	Status
Add Resource	Valid Data	Resource Added	Resource Added	PASS
Duplicate ID	Existing ID	Error Message	Error Displayed	PASS
Search Resource	ID = 101	Details Displayed	Details Displayed	PASS
Invalid Search	ID = 999	Not Found	Not Found	PASS
Resource Analysis	Qty < Minimum	Critical	Critical	PASS
Distribution	Valid Quantity	Stock Reduced	Stock Reduced	PASS
Excess Distribution	Qty > Stock	Insufficient Stock	Insufficient Stock	PASS
Sorting	Multiple Records	Sorted Data	Sorted Data	PASS
Save File	Valid Records	Data Stored	Data Stored	PASS
Generate Report	Resource Data	Report Created	Report Created	PASS

8. EXECUTION SCREENSHOTS / OUTPUT

The following screenshots should be inserted after executing the program in CodeBlocks, Dev C++, or VS Code.

8.1 Main Menu

SMART DISASTER RELIEF INVENTORY MANAGEMENT SYSTEM

1. Add Resource
2. Display All Resources
3. Update Resource
4. Search Resource by ID
5. Search Resource by Category
6. Sort Resources by Quantity
7. Sort Resources by Priority
8. Sort Resources by Distribution Centre
9. Analyse Resource Availability
10. Distribute Resource
11. Merge Resources
12. Delete Resource
13. Save Data to File
14. Load Data from File
15. Generate Disaster Resource Report
0. Exit

```
--- MAIN CONTROL PANEL ---
 1. View Inventory Dashboard & All Resources
 2. Add New Relief Resource Record
 3. Update Existing Resource Details
 4. Distribute / Allocate Relief Resources (Stock Outflow)
 5. Replenish / Restock Inventory (Stock Inflow)
 6. Delete / Decommission Resource Record
 7. Search Resources (Multi-Criteria & Recursive Binary Search)
 8. Sort Resources (Recursive Merge Sort with Multiple Keys)
 9. Multi-Centre Inventory Merge & Deduplication
10. Analytics, Critical Deficit & SDG Impact Reports
11. File Persistence & Report Generation
12. Load Sample Disaster Relief Simulation Dataset
 0. Exit System
```

```
--- MAIN CONTROL PANEL ---
 1. View Inventory Dashboard & All Resources
 2. Add New Relief Resource Record
 3. Update Existing Resource Details
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 9. Multi-Centre Inventory Merge & Deduplication
10. Analytics, Critical Deficit & SDG Impact Reports
11. File Persistence & Report Generation
12. Load Sample Disaster Relief Simulation Dataset
 0. Exit System
--------------------------
Enter your choice (0-12): 1

+------+----------+-----------------------+--------------+--------+--------+--------------------+-----------+----------+
| #    | ID       | Resource Name         | Category     | Avail. | MinReq | Relief Centre      | Priority  | Status   |
+------+----------+-----------------------+--------------+--------+--------+--------------------+-----------+----------+
| 1    | FOOD-001 | Ready-to-Eat Ration Pack | Food      |   1200 |   1500 | Zone-1 North Camp  | 5-Critical | LOW      |
| 2    | MED-002  | Paracetamol & Antibiotic | Medicine  |    320 |    300 | Zone-1 North Camp  | 3-High    | ADEQUATE |
| 3    | SHEL-002 | Tarpaulin Sheets (15x12 | Shelter    |    420 |    500 | Zone-1 North Camp  | 3-High    | LOW      |
| 4    | WATR-001 | Potable Drinking Water 2 | Water     |   1800 |   2000 | Zone-1 North Camp  | 5-Critical | LOW      |
| 5    | FOOD-002 | High-Energy Protein Bisc | Food      |    300 |    800 | Zone-2 Coastal Cam | 3-High    | CRITICAL |
| 6    | MED-001  | Emergency Trauma & First | Medicine  |     85 |    200 | Zone-2 Coastal Cam | 5-Critical | CRITICAL |
| 7    | MED-003  | Anti-Venom / Snakebite V | Medicine  |     15 |     50 | Zone-2 Coastal Cam | 5-Critical | CRITICAL |
| 8    | RESC-001 | Inflatable Life Rafts | Rescue       |     25 |     40 | Zone-2 Coastal Cam | 5-Critical | LOW      |
| 9    | RESC-002 | Life Jackets (Adult & Ch | Rescue    |    450 |    300 | Zone-2 Coastal Cam | 4-Urgent  | ADEQUATE |
| 10   | CLOT-001 | Thermal Fleece Blankets | Clothing   |   1100 |   1000 | Zone-3 East Camp   | 2-Medium  | ADEQUATE |
| 11   | SHEL-001 | Waterproof Family Tents | Shelter    |    150 |    400 | Zone-3 East Camp   | 4-Urgent  | CRITICAL |
| 12   | WATR-002 | Water Purification Table | Water     |   5000 |   2500 | Zone-3 East Camp   | 3-High    | ADEQUATE |
| 13   | 112      | first aid kit         | medicine     |  10000 |      5 | north              | 5-Critical | ADEQUATE |
+------+----------+-----------------------+--------------+--------+--------+--------------------+-----------+----------+
Total records: 13
```

8.2 Add Resource Output

Enter Resource ID: 101

Enter Resource Name: Food Packets

Enter Category: Food

Enter Quantity Available: 500

Enter Minimum Required Quantity: 300

Enter Distribution Centre: Chennai Central

Enter Priority Level: 1

Enter Distribution Demand: 250

```
Press [Enter] to return to menu...

--- MAIN CONTROL PANEL ---
 1. View Inventory Dashboard & All Resources
 2. Add New Relief Resource Record
 3. Update Existing Resource Details
 4. Distribute / Allocate Relief Resources (Stock Outflow)
 5. Replenish / Restock Inventory (Stock Inflow)
 6. Delete / Decommission Resource Record
 7. Search Resources (Multi-Criteria & Recursive Binary Search)
 8. Sort Resources (Recursive Merge Sort with Multiple Keys)
 9. Multi-Centre Inventory Merge & Deduplication
10. Analytics, Critical Deficit & SDG Impact Reports
11. File Persistence & Report Generation
12. Load Sample Disaster Relief Simulation Dataset
 0. Exit System
--------------------------
Enter your choice (0-12): 4

--- DISTRIBUTE RELIEF RESOURCES (STOCK OUTFLOW) ---
Enter Resource ID to distribute: FOOD-001
Resource: Ready-to-Eat Ration Packs | Current Available: 1200 units | Min Req: 1500 units
Enter Quantity to Distribute (> 0): 200

[SUCCESS] ALERT: Stock reached LOW level (1000 units remaining).

=======================================================
 RESOURCE CARD: Ready-to-Eat Ration Packs [FOOD-001]
=======================================================
  Resource ID       : FOOD-001
  Name              : Ready-to-Eat Ration Packs
  Category          : Food
  Quantity In-Stock: 1000 units
  Min Required      : 1500 units
  Stock Deficit     : 500 units
  Distribution Hub : Zone-1 North Camp
  Priority Level    : 5-Critical (5)
  Distributed Total: 650 units
  Health Status     : LOW
=======================================================
```

8.3 Resource Analysis Output

RESOURCE AVAILABILITY ANALYSIS

Resource: Food Packets
Status: LOW

Resource: Drinking Water
Status: CRITICAL
Immediate Replenishment Required!

```
 5. Replenish / Restock Inventory (Stock Inflow)
 6. Delete / Decommission Resource Record
 7. Search Resources (Multi-Criteria & Recursive Binary Search)
 8. Sort Resources (Recursive Merge Sort with Multiple Keys)
 9. Multi-Centre Inventory Merge & Deduplication
10. Analytics, Critical Deficit & SDG Impact Reports
11. File Persistence & Report Generation
12. Load Sample Disaster Relief Simulation Dataset
 0. Exit System
--------------------------
Enter your choice (0-12): 10

--- DISASTER RELIEF ANALYTICS & LOGISTICS AUDIT ---
 1. View Full Consolidated Analytics Dashboard (SDG Metrics & Stocks)
 2. View Critical Shortages & Deficit Analysis (Immediate Action)
 3. View Resources with Highest Distribution Demand (Top Surge)
 4. Generate Logistics Replenishment Advisory & Restock Plan
 0. Back to Main Menu
---------------------------------------------------
Select analytics option (0-4): 1

=========================================================================================
                  DISASTER RELIEF INVENTORY & HEALTH DASHBOARD
=========================================================================================
 Total Resource Types Registered : 14
 Total Physical Units Available  : 20865 units
 Total Relief Units Distributed  : 5250 units
-----------------------------------------------------------------------------------------
 STOCK HEALTH CLASSIFICATION:
   [!] CRITICAL Stock  :  4 items ( 28.6%) -> IMMEDIATE REPLENISHMENT REQUIRED!
   [*] LOW Stock       :  4 items ( 28.6%) -> MONITOR BUFFER LEVELS
   [V] ADEQUATE Stock  :  6 items ( 42.9%) -> SUFFICIENT OPERATIONAL BUFFER
-----------------------------------------------------------------------------------------
 SUSTAINABLE DEVELOPMENT GOAL (SDG) IMPACT METRICS:
   -> SDG 2: ZERO HUNGER (Food & Potable Water Available)     : 8100 units
   -> SDG 1 & 11: SUSTAINABLE COMMUNITIES & NO POVERTY        : 11465 units
      (Shelter, Trauma Kits, Medicine, and Rescue Equipment)
-----------------------------------------------------------------------------------------
 CENTRE-WISE STOCK & VULNERABILITY AUDIT:
 +---------------------------------+--------------+-------------+----------+------+
 | Distribution Centre             | Unique Items | Total Stock | Critical | Low  |
 +---------------------------------+--------------+-------------+----------+------+
 | Zone-1 North Camp               |            4 |        3540 |        0 |    3 |
 | Zone-2 Coastal Camp             |            5 |         875 |        3 |    1 |
 | Zone-3 East Camp                |            3 |        6250 |        1 |    0 |
 | north                           |            1 |       10000 |        0 |    0 |
 | south                           |            1 |         200 |        0 |    0 |
 +---------------------------------+--------------+-------------+----------+------+
 CATEGORY-WISE INVENTORY & DEMAND BREAKDOWN:
 +-----------------------+--------------+-------------+--------------------+
 | Category              | Unique Items | In-Stock    | Total Distributed |
 +-----------------------+--------------+-------------+--------------------+
 | Food                  |            2 |        1300 |               1350 |
 | Medicine              |            3 |         420 |                325 |
 | Shelter               |            2 |         570 |                630 |
 | Water                 |            2 |        6800 |               2150 |
 | Rescue                |            2 |         475 |                195 |
 | Clothing              |            1 |        1100 |                600 |
 | medicine              |            1 |       10000 |                  0 |
 | resuce                |            1 |         200 |                  0 |
 +-----------------------+--------------+-------------+--------------------+
```

```
--- DISASTER RELIEF ANALYTICS & LOGISTICS AUDIT ---
 1. View Full Consolidated Analytics Dashboard (SDG Metrics & Stocks)
 2. View Critical Shortages & Deficit Analysis (Immediate Action)
 3. View Resources with Highest Distribution Demand (Top Surge)
 4. Generate Logistics Replenishment Advisory & Restock Plan
 0. Back to Main Menu
---------------------------------------------------
Select analytics option (0-4): 2

==========================================================================================
              MOST CRITICAL SHORTAGES & DEFICIT ANALYSIS
==========================================================================================
+-----------+--------------------------+--------+--------+---------+--------------------+------------+
| ID        | Resource Name            | Avail. | MinReq | Deficit | Relief Centre      | Priority   |
+-----------+--------------------------+--------+--------+---------+--------------------+------------+
| FOOD-001  | Ready-to-Eat Ration Pack |   1000 |   1500 |     500 | Zone-1 North Camp  | 5-Critical |
| FOOD-002  | High-Energy Protein Bisc |    300 |    800 |     500 | Zone-2 Coastal Cam | 3-High     |
| SHEL-001  | Waterproof Family Tents  |    150 |    400 |     250 | Zone-3 East Camp   | 4-Urgent   |
| WATR-001  | Potable Drinking Water 2 |   1800 |   2000 |     200 | Zone-1 North Camp  | 5-Critical |
| MED-001   | Emergency Trauma & First |     85 |    200 |     115 | Zone-2 Coastal Cam | 5-Critical |
| SHEL-002  | Tarpaulin Sheets (15x12  |    420 |    500 |      80 | Zone-1 North Camp  | 3-High     |
| MED-003   | Anti-Venom / Snakebite V |     15 |     50 |      35 | Zone-2 Coastal Cam | 5-Critical |
| RESC-001  | Inflatable Life Rafts    |     25 |     40 |      15 | Zone-2 Coastal Cam | 5-Critical |
+-----------+--------------------------+--------+--------+---------+--------------------+------------+
 Total 8 items require priority stock replenishment.
```

```
Press [Enter] to return to menu...

--- DISASTER RELIEF ANALYTICS & LOGISTICS AUDIT ---
 1. View Full Consolidated Analytics Dashboard (SDG Metrics & Stocks)
 2. View Critical Shortages & Deficit Analysis (Immediate Action)
 3. View Resources with Highest Distribution Demand (Top Surge)
 4. Generate Logistics Replenishment Advisory & Restock Plan
 0. Back to Main Menu
---------------------------------------------------
Select analytics option (0-4): 4

==========================================================================================
              REPLENISHMENT ACTION PLAN & DISASTER LOGISTICS ADVISORY
==========================================================================================
 [TOP ACTION REQUIRED]:
 -> Single Most Critical Resource: Ready-to-Eat Ration Packs [FOOD-001]
    Centre: Zone-1 North Camp | Shortage: 500 units | Priority: 5-Critical
    Recommended Minimum Dispatch: 875 units immediately.

 General Restocking Recommendations:
 - [FOOD-001] Ready-to-Eat Ration Packs at Zone-1 North Camp: Current=1000, MinReq=1500 -> Order +875 units [LOW]
 - [SHEL-002] Tarpaulin Sheets (15x12 ft) at Zone-1 North Camp: Current=420, MinReq=500 -> Order +205 units [LOW]
 - [WATR-001] Potable Drinking Water 20L at Zone-1 North Camp: Current=1800, MinReq=2000 -> Order +700 units [LOW]
 - [FOOD-002] High-Energy Protein Biscuits at Zone-2 Coastal Camp: Current=300, MinReq=800 -> Dispatch +1300 units [CRITICAL]
 - [MED-001] Emergency Trauma & First Aid at Zone-2 Coastal Camp: Current=85, MinReq=200 -> Dispatch +315 units [CRITICAL]
 - [MED-003] Anti-Venom / Snakebite Vials at Zone-2 Coastal Camp: Current=15, MinReq=50 -> Dispatch +85 units [CRITICAL]
 - [RESC-001] Inflatable Life Rafts at Zone-2 Coastal Camp: Current=25, MinReq=40 -> Order +25 units [LOW]
 - [SHEL-001] Waterproof Family Tents at Zone-3 East Camp: Current=150, MinReq=400 -> Dispatch +650 units [CRITICAL]
==========================================================================================
```

9. ANALYSIS AND DISCUSSION

The developed Smart Disaster Relief Inventory and Distribution Management System provides a structured approach for managing essential resources during emergency situations. The use of an array of structures enables the storage of multiple resource records, while searching operations allow administrators to quickly locate required resources.

Sorting helps organize data based on quantity, priority, and distribution centre. The resource analysis module automatically identifies shortages by comparing available quantities with minimum required quantities and classifies resources as Adequate, Low, or Critical.

The merging functionality combines information from multiple relief centres while preventing duplicate records. Pointers are used for efficient resource access, while recursion is used for quantity analysis and identification of critical resources.

File handling provides permanent storage and allows data to be retrieved after program termination. Overall, the system demonstrates that fundamental and advanced C programming concepts can be combined to solve a practical disaster-management problem.

10. CONCLUSION

The Smart Disaster Relief Inventory and Distribution Management System using C was successfully designed to manage and monitor essential resources during disaster situations.

The system provides functionalities such as adding, updating, deleting, searching, sorting, merging, analysing, distributing, and storing resource information. The application successfully identifies resources that are Adequate, Low, or Critical and helps administrators identify shortages requiring immediate replenishment.

The project demonstrates important C programming concepts including operators, decision-making, loops, arrays, strings, structures, functions, searching, sorting, pointers, recursion, and file handling.

The system achieves the required Course Outcomes by applying programming concepts to a practical disaster-management problem. In the future, the project can be enhanced using databases, cloud storage, GPS tracking, IoT sensors, mobile applications, and artificial intelligence.

11. INDIVIDUAL CONTRIBUTION OF GROUP MEMBERS

The following table can be modified according to the actual project group members.

S.No	Name	Register Number	Contribution
1	Member 1	XXXXX	Problem analysis and system design
2	Member 2	XXXXX	Resource management and searching
3	Member 3	XXXXX	Sorting, analysis and merging
4	Member 4	XXXXX	File handling, testing and documentation

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